

Lucas Fernandez Sarmiento

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Professional Summary

PhD researcher in theoretical physics and machine learning at Carnegie Mellon, using statistical mechanics, probabilistic modeling, and large-scale Python experiments to study high-dimensional learning systems. Recipient of a 2026–2027 CMU Presidential Scholarship, a Mellon College of Science-wide award funded by a Tata Consultancy Services (TCS) gift. Single-author ICML 2026 paper on dropout scheduling and neural-network training dynamics. Experience building forecasting, Monte Carlo simulation, optimization, backtesting, and robust-aggregation pipelines with PyTorch, scikit-learn, and pandas/NumPy.

Education

Carnegie Mellon University PhD Candidate and MS, Theoretical Physics (GPA: 4.00/4.00)	<i>2022–2027 (expected)</i>
Imperial College London MSc, Theoretical Physics (Distinction)	<i>2021–2022</i>
Imperial College London BSc, Theoretical Physics (First Class, Dean's List)	<i>2018–2021</i>

Research and Professional Experience

PhD Researcher <i>Carnegie Mellon University</i> <i>Advisors: Riccardo Penco and Rachel A. Rosen</i>	<i>Pittsburgh, PA</i> <i>Aug 2022–present</i>
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- Derived a mean-field theory of dropout in stochastic neural-network dynamics, predicting scaling laws and depth-dependent schedules. Across MLP, ResNet, and ViT models on CIFAR-10/100, improved accuracy, lowered test loss by 18–35%, and reduced compute by up to 65% at matched accuracy.
- Built robust aggregation experiments for adversarial LLM debate under asymmetric information, comparing majority vote, linear pooling, Hedge/logarithmic pooling, and cost-accuracy Pareto frontiers; exceeded state-of-the-art single-model performance at about 70% lower inference cost.
- Built Python workflows for large-scale tensor-network simulations with Prof. Tarnopolsky, running DMRG experiments with scaling analysis and convergence diagnostics for lattice-model observables.
- Extended mean-field and signal-propagation methods to residual networks, studying depth-dependent residual scaling and improved He initialization for stable training at large depth.
- Developed symmetry-based effective field theory constructions using coset methods, tensor calculus, and algebraic geometry; applied these tools to gravitational EFTs with scale symmetry, identifying connections across field theory and gravity.

Adjunct Lecturer <i>Universidad de las Hespérides</i>	<i>Madrid, Spain</i> <i>Apr 2025–Dec 2026</i>
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- Designed and taught quantitative-finance mathematics modules covering linear algebra, differential calculus, ODE/PDE methods, and constrained optimization, with applications to portfolio-risk optimization.
- Developed Spanish ML lecture notes on data matrices, probability, PCA, SVD, least squares, Ridge/Lasso, logistic regression, cross-validation, and backprop for quant-finance examples.

Co-founder and Technical Lead <i>SteamAI</i>	<i>Luxembourg</i> <i>Jun 2024–Aug 2024</i>
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- Built multivariate time-series forecasting and constrained-control models for refinery steam networks under stochastic demand, combining autoregressive demand modelling with pressure-constrained optimization to reduce vented steam.

Awards & Honors

2026–2027	CMU Presidential Fellowship, Mellon College of Science, funded by Tata Consultancy Services
2026	NSF ACCESS Compute Allocation, 200,000 ACCESS credits (~2,000 H100 GPU-hours; PI)
2025	CMU Professional Engagement Award
2023	Rafael del Pino Foundation Excellence Scholarship (\$28,600)
2021	UROP Bursary Prize, Imperial College London (~\$3,000)
2019, 2020	Dean's List, Imperial College London
2018	Finalist, Big Bang Young Scientist of the Year (UK)
2017	Gold CREST Award (Jaguar Land Rover research)

Selected Projects

- LaLiga simulations: Built an end-to-end forecasting and backtesting pipeline using Understat xG/match data, pandas/NumPy feature engineering, scikit-learn score models, and Monte Carlo season simulations to compare expected-points forecasts with bookmaker-implied probabilities.
- English-to-French translation: Trained GRU, LSTM, and attention-based Transformer seq2seq models in PyTorch and fine-tuned Llama 2 7B with QLoRA on WMT 2015, improving BLEU from 1.6 to 13.6.

Technical Skills

Quantitative Modeling: Time-series forecasting, Monte Carlo simulation, stochastic processes, probabilistic modeling, backtesting, calibration, out-of-sample validation, bookmaker-implied probabilities, credit-risk scoring.

Python / ML Stack: Python, pandas, NumPy, scikit-learn, PyTorch, Hugging Face, asyncio, Git, AWS.

Optimization / Decision Systems: Convex optimization, constrained optimization, projected gradient methods, no-regret learning, probabilistic opinion pooling, game theory, Markov games, MDPs, PPO, multi-agent RL.

Deep Learning / Scientific Computing: Transformers, ResNets, ViTs, LSTMs, CNNs, LoRA/QLoRA, tensor networks, DMRG, SLURM/HPC cluster computing, Mathematica, MATLAB, LaTeX.

Coursework: CMU ML (10-701), CMU AI Agents (11-768), CMU Adv. ML & Game Theory (17-759).

Selected Publications and Manuscripts

- Lucas Fernández-Sarmiento. *Dropout Universality: Scaling Laws and Optimal Scheduling at the Edge-of-Chaos* [🔗](#). Single-author paper accepted to ICML 2026; develops a predictive theory of dropout, regime changes in neural-network dynamics, and improved generalization through optimal scheduling.
- Keltin Grimes, Lucas Fernández-Sarmiento. *Don't Debate, Interrogate! Robust AI Debate Despite Deceptive Agents* (under preparation). Studies robust aggregation under correlated, non-IID judge errors.
- Lucas Fernández-Sarmiento, Sophie Merchant, Rupert Mitchell. *Mean-Field Theory for ResNets: Depth-Dependent Residual Connections and Improved He Initialization* (under preparation).
- Lucas Fernández-Sarmiento, Riccardo Penco, Rachel A. Rosen. *On Entropy Bounds for Irrelevant Operators*. Accepted to JHEP. [arXiv:2508.14978](#) [🔗](#).
- Lucas Fernández Sarmiento, Irvin Martínez Rodríguez. *Broken Weyl Gravity: Dynamical Torsion and Generalized Proca*. [arXiv:2510.08778](#) [🔗](#).

Conference Presentations

- Scheduled to attend and present at *ICML 2026*.
- Expected presenter, *IAIFI Summer Workshop 2026* (upcoming; abstract submitted): *Dropout Universality: Scaling Laws and Optimal Scheduling at the Edge-of-Chaos*.
- Presented physics research at COSMO 2025 and the Cargèse BSM conference/summer school; attended *When QFT Meets QI*, *Contours 2026*, and *Amplitudes London 2026*.